

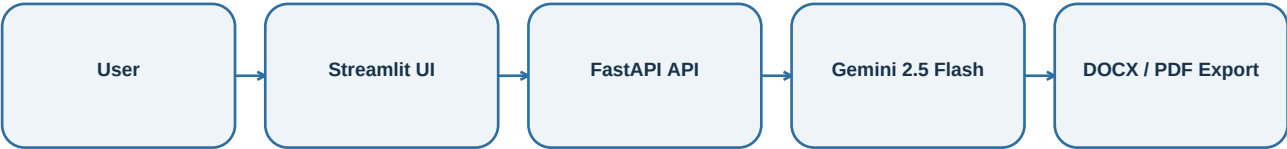
LegalEase: AI-Powered Legal Document Generator

Technology Stack (Architecture & Stack)

Project Phase	2. Requirement Analysis
Team	Team 3
Date	02 October 2026
Prepared From	Supplied LegalEase repository + supplied project templates

Technical architecture

Current implementation flow



Components & technologies

S.No.	Component	Description	Technology
1	User Interface	Form, preview and downloads	Streamlit + CSS
2	Application Logic	Validation and generation endpoint	FastAPI + Pydantic
3	AI Generation	Draft legal-style text	Google Gemini API / google-generativeai
4	Formatting	DOCX/PDF/HTML rendering	python-docx + fpdf2
5	HTTP Client	Frontend-to-backend calls	requests
6	Configuration	API key, model and URL	python-dotenv
7	Testing	Automated API and formatter tests	pytest + TestClient
8	Database	Persistent database	Not implemented
9	Infrastructure	Local Python/uvicorn run	Python 3.10+

Application characteristics

Characteristic	Current implementation
Open-source frameworks	FastAPI, Streamlit, python-docx, fpdf2, pytest
Security	Secret API key is expected through environment configuration; .env is not committed.
Scalability	UI and API are separated and can be deployed independently later.
Availability	No load balancer or HA configuration is present.

Characteristic	Current implementation
Performance	90-second frontend request timeout; Gemini retries up to three attempts.

Source basis: Implemented claims are based on the supplied LegalEase repository. Items without repository evidence are explicitly marked pending, future, or not implemented.